

## Project Workflow: Week 2

### Phase: Requirements Analysis & System Design

#### 1. Objective

The primary goal of Week 2 was to plan the internal logic and structure of the game to ensure the code remains organized, modular, and easy to maintain throughout the development process.

#### 2. Detailed Task Breakdown

- **Requirement Finalization:** Defined key game mechanics including score limits, paddle control logic, and ball collision boundaries to ensure a consistent gameplay experience.
- **Architecture Strategy:** Established a plan to separate the game's visual display from the background calculations, ensuring that rendering tasks and game physics are managed independently.
- **Logic Structuring:** Established the operational flow for different game stages (Start, Playing, Game Over) to ensure the code handles user inputs and game events smoothly.

#### 3. Deliverables

- **Functional Requirements Summary:** A documented list of game mechanics and rules.
- **Modular Architecture Plan:** A structured plan defining how different parts of the code (Paddle, Ball, Score) will interact without overlapping responsibilities.
- **Game Logic Workflow:** A defined sequence of operations for handling user input and real-time updates.

#### 4. Tools & Resources

- **Development Strategy:** Object-Oriented Programming (OOP) principles to ensure code modularity.
- **Documentation:** Project log for tracking requirements and logic updates.

Planned Class Structure

| Class Name   | Primary Responsibility                                                         |
|--------------|--------------------------------------------------------------------------------|
| GamePanel    | Acts as the main game screen; manages rendering and the timer-based game loop. |
| Ball         | Controls ball movement, speed, and collision physics against walls.            |
| Paddle       | Manages player input for vertical movement and paddle positioning.             |
| ScoreManager | Tracks scores and triggers the win/loss condition once a limit is reached.     |
| GameLogic    | Handles the interaction between the ball and paddles to update the game state. |